Answer Key & Teacher Notes

Session 1 — Finding Mean Absolute Deviation (MAD) · Session 2 — Comparing Data Sets Using MAD

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Finding Mean Absolute Deviation (MAD)

TALK PROMPTS

- Why do you use distance from the mean instead of whether a score is above or below it?
- How would the total of the distances change if one evening moved farther from the mean?

PARTNER TASK

Gym Check. One partner finds the mean of the gym visitors in Problem 2 (31, 35, 41, 45; mean 38); the other finds each distance and the MAD (5). Switch with Group 2 in Problem 8 (4, 9, 10, 13; mean 9, MAD 2.5).

WATCH FOR

- Adding signed differences so they cancel to 0 instead of using distances.
- Dividing the sum of distances by the mean instead of by the number of values.
- Leaving out a 0 distance when averaging.

SESSION 2 · TEACHER NOTES

Comparing Data Sets Using MAD

TALK PROMPTS

- Two shops share a mean. What does the MAD add that the mean cannot say by itself?
- How can two groups with the very same mean feel completely different?
- What would a MAD of zero tell you about a class?

PARTNER TASK

Garden Beds. In Problem 5, one partner finds Bed 1's MAD (6 cm); the other finds Bed 2's MAD (11 cm). Together decide which bed grew more evenly and why.

WATCH FOR

- Thinking a larger mean means a larger MAD (center and spread are different measures).
- Choosing the set with the larger MAD as the more consistent one.
- Concluding that equal means (or equal MADs) make two data sets identical.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.5c	1	2	2	2	2	2	2	2	2	1	1
S1 · Form B	6.SP.B.5c	1	2	2	2	2	2	2	2	2	1	1
S2 · Form A	6.SP.B.5c	2	2	2	2	2	2	3	2	2	1	1
S2 · Form B	6.SP.B.5c	2	2	2	2	2	2	3	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

mean absolute deviation (MAD): The **mean** of the distances from the mean.

deviation: How **far** one value is from the mean.

absolute value: A number's distance from 0; it is never **negative**.

spread: How far **apart** the values are.

- Find the mean, then each value's **distance** from the mean.
- The MAD is the **mean** of those distances.
- A small MAD means the values are **close** to the mean.

Example: Mean: 28 ÷ 4 = **7** Distances from 7: 4, **0**, 1, 3 Add the distances: **8**. Divide by 4. Answer: MAD = **2** cm

YOU TRY

1. a. (3 + 1 + 1 + 3) ÷ 4 = 2 b. (7 + 1 + 1 + 7) ÷ 4 = 4

2. distances 4, 2, 0, 2, 4; sum 12; 12 ÷ 5 = 2.4

EXIT TICKET

1. a. 8 ÷ 4 = 2 b. 2 ÷ 4 = 0.5

2. Use distances 4, 1, 2, 3: sum 10; MAD = 10 ÷ 4 = 2.5

3. **B** (A typical value is about 4 from 72.)

PRACTICE FORM A

1. Distances: 6, 4, 0, 4, 6. Sum = 20. MAD = 20 ÷ 5 = 4

2. a. 152 ÷ 4 = 38 visitors b. distances 7, 3, 3, 7; sum 20; MAD = 20 ÷ 4 = 5 visitors

3. mean 58; distances 5, 1, 1, 5; sum 12; MAD = 12 ÷ 4 = 3°F

4. No. Mean 25; every distance is 0, so MAD = 0

5. A typical delivery is about 3 minutes from 45 minutes (roughly 42 to 48 minutes).

6. mean 100 ÷ 5 = 20; distances 8, 2, 0, 2, 8; sum 20; MAD = 4 pages

7. Total of distances = 5 × 4 = 20; 20 − (6 + 2 + 4) = 8

8. a. distances 3, 1, 1, 3; MAD = 8 ÷ 4 = 2 b. distances 5, 0, 1, 4; MAD = 10 ÷ 4 = 2.5 c. The means match, but the MAD shows Group 1's makes sit closer to 9.

9. The signs cancel. Distances are never negative, so add 5 + 3 + 3 + 5 = 16. MAD = 16 ÷ 4 = 4.

10. **D** (mean absolute deviation)

11. **B** (negative)

Self-check word **DEVIATE**

PRACTICE FORM B

1. Distances: 7, 4, 0, 4, 7. Sum = 22. MAD = 22 ÷ 5 = 4.4

2. a. 160 ÷ 4 = 40 visitors b. distances 8, 4, 4, 8; sum 24; MAD = 24 ÷ 4 = 6 visitors

3. mean 66; distances 5, 2, 2, 5; sum 14; MAD = 14 ÷ 4 = 3.5°F

4. No. Mean 40; every distance is 0, so MAD = 0

5. A typical ride is about 2 minutes from 30 minutes (roughly 28 to 32 minutes).

6. mean 100 ÷ 5 = 20; distances 10, 4, 2, 4, 8; sum 28; MAD = 28 ÷ 5 = 5.6 pages

7. Total of distances = 6 × 4 = 24; 24 − (3 + 7 + 5) = 9

8. a. distances 3, 1, 1, 3; MAD = 8 ÷ 4 = 2 b. distances 6, 0, 1, 5; MAD = 12 ÷ 4 = 3 c. The means match, but the MAD shows Group 1's makes sit closer to 8.

9. The signs cancel. Distances are never negative, so add 5 + 3 + 3 + 5 = 16. MAD = 16 ÷ 4 = 4.

10. **A** (mean)

11. **C** (closer to the mean)

Self-check word **AVERAGE**

GUIDED NOTES

variation: How **different** the values are from each other.

consistent: Values stay **close** together; the spread is small.

MAD: The mean **distance** of the values from the mean.

- Two sets can have the same mean and **different** MADs.
- The set with the **smaller** MAD is steadier (more consistent).

Example: A's distances: 2, 1, 1, 2. MAD = 6 ÷ 4 = **1.5** B's distances: 6, 2, 2, 6. MAD = 16 ÷ 4 = **4** Answer: Baker **A** is steadier: 1.5 < 4.

YOU TRY

1. a. 12 ÷ 4 = 3 b. 32 ÷ 4 = 8 (Set 1 is more consistent)

2. Machine A (smaller MAD: 2 < 14)

EXIT TICKET

1. a. 8 ÷ 4 = 2 b. 20 ÷ 4 = 5

2. Set M (MAD 1.5 < 7, so its values stay closer to 25)

3. **B** (Their spreads may differ.)

PRACTICE FORM A

1. a. distances 6, 2, 2, 6; MAD = 16 ÷ 4 = 4 b. distances 15, 3, 3, 15; MAD = 36 ÷ 4 = 9 c. Shop 1 (4 < 9)

2. a. distances 7, 3, 3, 7; MAD = 20 ÷ 4 = 5 min b. distances 6, 2, 2, 6; MAD = 16 ÷ 4 = 4 min c. Trail B (smaller MAD)

3. No. They have the same spread (MAD 3) but different centers (90 and 45).

4. Team N: same mean, but its scores stay about 3 points from 82 (MAD 3 < 14).

5. a. distances 9, 3, 3, 9; MAD = 24 ÷ 4 = 6 cm b. distances 18, 4, 4, 18; MAD = 44 ÷ 4 = 11 cm c. Bed 1

6. a. distances 10, 4, 4, 10; MAD = 28 ÷ 4 = 7 b. distances 7, 7, 7, 7; MAD = 7 c. Same center and same typical distance, but the values are arranged differently (K has two pairs; J is spread evenly).

7. Supplier 1: one day slower on average, but a MAD of 0.5 day makes it predictable; Supplier 2 can be days late.

8. a. distances 1, 0, 1, 0; MAD = 2 ÷ 4 = 0.5 b. distances 4, 2, 2, 4; MAD = 12 ÷ 4 = 3 c. Archer 1: same mean, but scores stay close to 8.

9. The mean tells the center and the MAD tells the spread; they answer different questions. She must compare the MADs. Example: 49, 50, 50, 51 (mean 50, MAD 0.5) and 2, 6, 14, 18 (mean 10, MAD 6).

10. **B** (smaller MAD)

11. **A** (Set A)

Self-check word **DISTANCE**

PRACTICE FORM B

1. a. distances 6, 2, 2, 6; MAD = 16 ÷ 4 = 4 b. distances 18, 4, 4, 18; MAD = 44 ÷ 4 = 11 c. Stand 1 (4 < 11)

2. a. distances 8, 3, 3, 8; MAD = 22 ÷ 4 = 5.5 min b. distances 5, 2, 2, 5; MAD = 14 ÷ 4 = 3.5 min c. Trail B (smaller MAD)

3. No. They have the same spread (MAD 4) but different centers (80 and 40).

4. Team Q: same mean, but its scores stay about 2 points from 75 (MAD 2 < 12).

5. a. distances 7, 3, 3, 7; MAD = 20 ÷ 4 = 5 cm b. distances 16, 4, 4, 16; MAD = 40 ÷ 4 = 10 cm c. Bed 1

6. a. distances 8, 4, 4, 8; MAD = 24 ÷ 4 = 6 b. distances 6, 6, 6, 6; MAD = 6 c. Same center and same typical distance, but the values are arranged differently (K has two pairs; J is spread evenly).

7. Supplier 1: one day slower on average, but a MAD of 0.5 day makes it predictable; Supplier 2 can be days late.

8. a. distances 1, 1, 1, 1; MAD = 4 ÷ 4 = 1 b. distances 4, 3, 3, 4; MAD = 14 ÷ 4 = 3.5 c. Archer 1: same mean, but scores stay close to 7.

9. The mean tells the center and the MAD tells the spread; they answer different questions. He must compare the MADs. Example: 99, 100, 100, 101 (mean 100, MAD 0.5) and 4, 8, 16, 20 (mean 12, MAD 6).

10. **B** (spread)

11. **C** (mean and MAD)

Self-check word **ABSOLUTE**

SHEET 1 · FINDING MEAN ABSOLUTE DEVIATION (MAD)

1. Distances: 5, 3, 0, 2, 6. Sum = 16. MAD = 16 ÷ 5 = 3.2

2. a. (3 + 1 + 1 + 3) ÷ 4 = 2 b. (10 + 5 + 5 + 10) ÷ 4 = 7.5 c. 0 (every value equals the mean) d. (5 + 2 + 1 + 2 + 4) ÷ 5 = 2.8

3. a. 16 ÷ 8 = 2 pets b. distances 2, 1, 1, 0, 0, 1, 1, 2; sum 8; MAD = 8 ÷ 8 = 1 pet

4. a. 96 ÷ 6 = 16 min b. distances 6, 2, 1, 1, 2, 6; sum 18; MAD = 18 ÷ 6 = 3 min (a typical wait is about 3 minutes from 16)

5. He divided by the mean (8) instead of the number of values (5). MAD = 16 ÷ 5 = 3.2

SHEET 2 · COMPARING DATA SETS USING MAD

1. a. (2 + 1 + 0 + 1 + 2) ÷ 5 = 1.2 points b. (8 + 4 + 0 + 4 + 8) ÷ 5 = 4.8 points c. Kai (smaller MAD)

2. a. 48 ÷ 8 = 6 books b. distances 4, 3, 2, 0, 0, 2, 3, 4; sum 18; MAD = 18 ÷ 8 = 2.25 books c. Club B (MAD 2.25 versus 0.5)

3. a. distances 2, 1, 1, 2; MAD = 6 ÷ 4 = 1.5 tomatoes b. distances 8, 4, 4, 8; MAD = 24 ÷ 4 = 6 tomatoes c. Garden A (1.5 < 6)

4. Route 5: its smaller MAD means its times stay closer to 24 minutes.

5. Set R: mean 50, distances 1, 0, 0, 1, MAD 0.5. Set S: mean 10, distances 8, 4, 4, 8, MAD 6. Jordan is wrong: bigger values do not mean more spread; Set S is more spread out.

6. **A** (A varies less.)

7. **B** (2)

October 2026 conceptual additions — teacher key

Session 1 · Finding Mean Absolute Deviation (MAD)

Mean 5 cm; distances 3, 1, 1, 3 cm; total distance 8 cm; MAD 2 cm. Distances are nonnegative; using signed differences would cancel and would not measure typical distance.

Extension: The mean locates the center at 5 cm, while the MAD says observations are an average distance of 2 cm from that center.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Comparing Data Sets Using MAD

Both means are 5. A distances 1,0,0,1: MAD 0.5. B distances 4,2,2,4: MAD 3. A is more tightly clustered around its mean.

Extension: 5,5,5,5; all distances from 5 are zero.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.